

objects. When a local variable is used in the source code, the variable with that name in the topmost frame object is used.

Every running program has a call stack, and multithreaded programs have one call stack for each thread. But when you look at the source code for a program, you can't see the call stack in the code. The call stack isn't stored in a variable as other data structures are; it's automatically handled in the background.

The fact that the call stack doesn't exist in source code is the main reason recursion is so confusing to beginners: recursion relies on something the programmer can't even see! Revealing how stack data structures and the call stack work removes much of the mystery behind recursion. Functions and stacks are both simple concepts, and we can use them together to understand how recursion works.